$$I = \int_0^r j 2\pi r dr = E \int_0^r \sigma_0 2\pi r dr = \frac{2\pi R^3 \beta E}{3}. \quad (8)$$

$$E = \frac{3I}{2\pi\beta R^3}. \quad (9)$$

$$P_r = \int_0^r \sigma_0 E^2 2\pi r L dr = \frac{3I^2 L r^3}{2\pi\beta R^6}. \quad (10)$$

$$P_r = P = -\kappa 2\pi r L \frac{dT}{dr}. \quad (11)$$

(11) (10)

$$T(R) = T_s, \quad (12)$$

$$T(r) = T_0 + \frac{I^2(\alpha R^3 + 3\kappa R^2 - \alpha r^3)}{4\pi^2 \alpha\beta\kappa R^6}. \quad (13)$$

$$T_{\max} = T_0 + \frac{I^2(\alpha R^3 + 3\kappa R^2)}{4\pi^2 \alpha\beta\kappa R^6} = 299K. \quad (14)$$

4.

$$\delta R_T = \int_0^R \gamma \left[T(r) - T_0\right] dr = \frac{3\gamma(\alpha R + 4\kappa) I^2}{16\pi^2 \alpha\beta\kappa R^3} = 5.70 \times 10^{-9} m. \quad (15)$$

5. .

$$B 2\pi r = \int_0^r j 2\pi r dr = E \int_0^r \sigma_0 2\pi r dr. \quad (16)$$

(9)

$$B(r) = \frac{\mu_0 I r^2}{2\pi R^3}. \quad (17)$$

6.

$$w_B(r) = \frac{B^2(r)}{2\mu_0}, \quad (18)$$

$$W_B = \int_0^R w_B(r) 2\pi r L dr = \frac{\mu_0 I^2 L}{24\pi} = 8.33 \times 10^{-10} J. \quad (19)$$

7. , l L .

$r, r + dr.$

$$dF_A = jB(r) L l dr. \quad (20)$$

$$dp(r) = \frac{dF_A}{lL} = \frac{3\mu_0 I^2 r^3}{4\pi^2 R^6} dr. \quad (21)$$